# Wk9L1: 2nd Order Circuits

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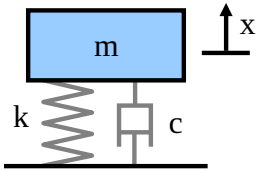


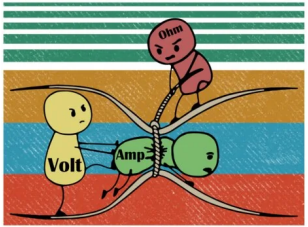


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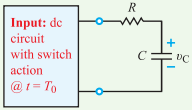
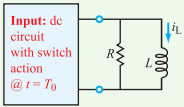
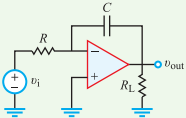
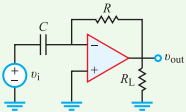
**Table 5-5:** Response forms of basic first-order circuits.

| Circuit | Diagram | Response |
|---|---|---|
| RC |  | $v_C(t) = \left\{ v_C(\infty) + [v_C(T_0) - v_C(\infty)]e^{-(t-T_0)/\tau} \right\} u(t - T_0)$ <br> $(\tau = RC)$ |
| RL |  | $i_L(t) = \left\{ i_L(\infty) + [i_L(T_0) - i_L(\infty)]e^{-(t-T_0)/\tau} \right\} u(t - T_0)$ <br> $(\tau = L/R)$ |
| Ideal integrator |  | $v_{out}(t) = -\frac{1}{RC}\int_{t_0}^{t} v_i \, dt' + v_{out}(t_0)$ |
| Ideal differentiator |  | $v_{out}(t) = -RC\frac{dv_i}{dt}$ |

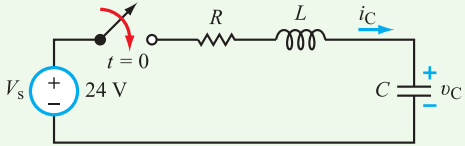


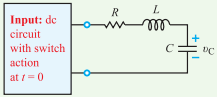
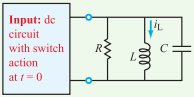

**Figure 6-7:** Series RLC circuit connected to a source $V_s$ at $t = 0$. In general, the capacitor may have had an initial charge on it at $t = 0^-$, with a corresponding initial voltage $v_C(0^-)$.

Circuit simulator: www.falstad.com/circuit/

Load `Wk9L1_RLC1.circuitjs.txt`

- Overdamped:
  $\alpha > \omega_0 \Rightarrow R > 2\sqrt{L/C}$
- Critically Damped:
  $\alpha = \omega_0 \Rightarrow R = 2\sqrt{L/C}$
- Underdamped:
  $\alpha < \omega_0 \Rightarrow R < 2\sqrt{L/C}$

**Table 6-1: Step response of RLC circuits for $t \geq 0$.**

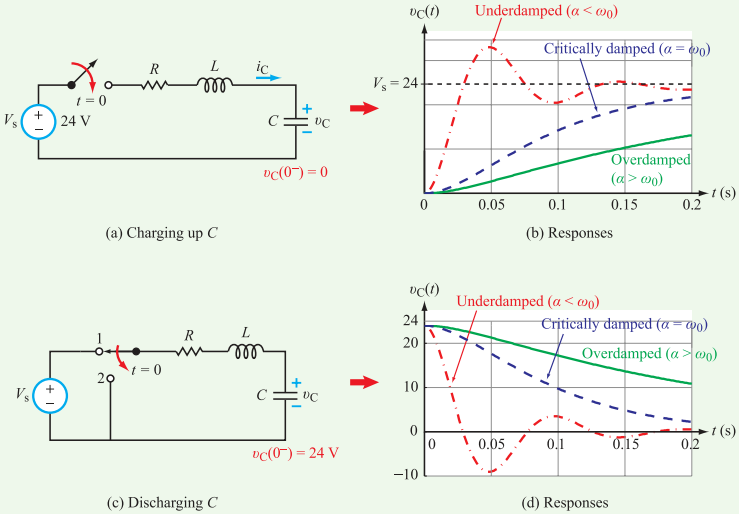
| Series RLC | Parallel RLC |
|---|---|
|   |   |
| **Total Response** | **Total Response** |
| **Overdamped** ($\alpha > \omega_0$)<br>$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$<br>$A_1 = \dfrac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$<br>$A_2 = \dfrac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$ | **Overdamped** ($\alpha > \omega_0$)<br>$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$<br>$A_1 = \dfrac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$<br>$A_2 = \dfrac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$ |
| **Critically Damped** ($\alpha = \omega_0$)<br>$v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$<br>$B_1 = v_C(0) - v_C(\infty)$<br>$B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$ | **Critically Damped** ($\alpha = \omega_0$)<br>$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$<br>$B_1 = i_L(0) - i_L(\infty)$<br>$B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$ |
| **Underdamped** ($\alpha < \omega_0$)<br>$v_C(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$<br>$D_1 = v_C(0) - v_C(\infty)$<br>$D_2 = \dfrac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$ | **Underdamped** ($\alpha < \omega_0$)<br>$i_L(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$<br>$D_1 = i_L(0) - i_L(\infty)$<br>$D_2 = \dfrac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$ |

**Auxiliary Relations**

$$\alpha = \begin{cases} \dfrac{R}{2L} & \text{Series RLC} \\ \dfrac{1}{2RC} & \text{Parallel RLC} \end{cases} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$




(a) Charging up $C$

(b) Responses

(c) Discharging $C$

(d) Responses

**Figure 6-4:** Illustrating the charge-up and discharge responses of a series RLC circuit with $V_S = 24$ V. In all cases $R = 12\ \Omega$ and $L = 0.3$ H, which specifies $\alpha = R/2L = 20$ Np/s. When $C = 0.01$ F, the response is overdamped, when $C = 8.33$ mF, the response is critically damped, and when $C = 0.72$ mF, the response is underdamped.

# Reminders

- HW 6 Due
- HW 7 Posted
- Demo Wednesday